



CARIBBEAN EXAMINATIONS COUNCIL
ADVANCED PROFICIENCY EXAMINATION

ACCOUNTING

UNIT 2 – PAPER 02

2 ¾ hours

01 JUNE 2010 (p.m.)

1. This paper consists of **THREE** questions.
2. **EACH** question is worth 35 marks.
3. **ALL** questions are **COMPULSORY**.
4. Begin **EACH** answer on a new page.
5. Silent non-programmable calculators may be used, but **ALL** necessary working should be clearly shown.



1. (a) Below are listed various costs that might be found in a service, merchandising, or manufacturing company.

1. Bread used to make sandwiches in a cafeteria
2. Advertising by an engineering firm
3. Mangoes processed and canned by an agricultural cooperative
4. Shipping of canned mangoes for an agricultural cooperative to its customers
5. Insurance on a factory producing carnival costumes
6. Insurance on the corporate headquarters of a leading Caribbean Insurance Company
7. Salary of a supervisor overseeing production of handicraft items
8. Commissions paid to persons selling advertising space in a local newspaper
9. Depreciation on staff lunchroom facilities at a leading manufacturing firm
10. Steering wheels used by a company in the production of automobiles
11. Property taxes on a factory building
12. Salaries of janitorial staff assigned to the administrative offices
13. Lubricants used on printing equipment in a print shop
14. Solder used in producing electrical appliances
15. Bad debt expenses
16. Boxes used in packaging soap powder

- (i) Classify EACH cost as being **either** variable **or** fixed with respect to volume or level of activity.
- (ii) Classify EACH cost as being **either** a selling and administrative cost **or** a product cost.

Prepare your responses, using the following format. The first item has been completed as an example.

Cost Variable	Cost Behaviour		Selling and Administrative	Product Cost
	Fixed	Cost		
1.	X			X

[15 marks]

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- (b) Darren Joseph is employed by Arima Products and assembles a component part for one of the company's product lines. He is paid \$20 per hour for regular time, and time and a half for all work in excess of 40 hours per week.
- (i) Assume that during a given week Darren is idle for five hours due to machine breakdowns and that he is idle for four more hours due to material shortages. No overtime is recorded for the week. Allocate Darren's wages for the week between direct labour cost and manufacturing cost. **[3 marks]**
 - (ii) Assume that during the following week Darren works a total of 48 hours. He has no idle time for the week. Allocate Darren's wages for the week between direct labour cost and manufacturing overhead cost. **[3 marks]**
 - (iii) Arima Products provides an attractive package of fringe benefits for its employees. This package includes a retirement programme and a health insurance programme. Explain TWO ways that the company could classify the cost of fringe benefits for direct labour workers in its cost records. **[4 marks]**
- (c) You have gathered the following information relating to Montego Bay Company, which has two service departments and two production departments:

	Service Department		Production Department	
	1	2	A	B
Direct overhead costs	\$268 000	\$228 000	\$150 000	\$240 000
Square feet occupied	15 000	10 000	10 000	13 000
Number of employees	15	10	48	28
Cost allocation basis	Square Feet	Number of Employees		

Montego Bay Company makes no distinction between fixed and variable costs incurred in its service departments.

Calculate the total service department costs that should be allocated to EACH production department using the direct method. **[10 marks]**

Total 35 marks

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NOTHING HAS BEEN OMITTED

2. (a) The Regional Printing Company produces phone books for firms located throughout the Eastern Caribbean. A job order costing system is used to accumulate the costs incurred for each printing job. At the beginning of August 2009, two jobs were in process: Job 811 with assigned costs of \$16 500 and Job 813 with assigned costs of \$4 200. The following transactions were recorded by the Company during August 2009:

- Raw materials costing \$69 500 were purchased on account.
- Shop supplies costing \$3 600 were purchased on account.

Raw materials and shop supplies requisitioned during August were:

Raw Materials:

Job 813	\$ 8 500
Job 814	16 500
Job 815	12 000
Shop supplies	3 300

Labour hours worked during the month were:

Job 811	410
Job 813	620
Job 814	750
Job 815	110
Indirect labour	140

- Each worker was paid \$14 per hour.
- Depreciation on shop equipment was \$7 200.
- Other shop overhead was \$9 150.
- Shop overhead was applied to printing jobs at the rate of \$13.20 per direct labour hour.
- Jobs 811, 813 and 814 were completed and sold during the month.

- (i) Calculate the total shop overhead applied to jobs in August. [2 marks]
- (ii) By how much was shop overhead over- or under-applied during August? [3 marks]
- (iii) Compute the cost of the completed jobs for the month. [10 marks]
- (iv) Compute the total cost of jobs in process on August 31. [4 marks]

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- (b) (i) Describe the MAIN difference between traditional costing systems and activity-based costing systems. [2 marks]

- (ii) An Antigua Company manufactures two products, premium and super. The company has manufactured the super product since its inception; the premium product was introduced several years later to tap into a new segment of the market. Since the introduction of the premium product, the company's profits have steadily declined, and management has become concerned about the accuracy of its costing system. Sales of the premium product have been increasing rapidly.

Overhead is currently assigned to the products on the basis of direct labour hours, and the company's predetermined overhead rate for 2009 is \$25 per direct labour hour.

For 2009, the current year, the company has estimated that it will incur \$4 000 000 in overhead costs and produce 10 000 units of premium and 80 000 units of super. The premium product requires 3.2 hours of direct labour time per unit, and the super product requires 1.6 hours per unit. Direct materials, direct labour and overhead costs per unit are as follows.

	Products	
	Premium	Super
Direct materials	\$300	\$224
Direct labour	32	26
Manufacturing overhead	80	40
Total manufacturing cost	\$412	\$290

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Assume that the company's overhead costs can be traced to four activity centres. These activity centres, their cost drivers, and estimated cost and activity data for each centre for 2009 are given below:

Activity Centre and (Cost Driver)	Traceable Cost	Expected Number of Transactions or Events		
		Premium	Super	Total
Purchase orders (Numbers of orders)	\$168 000	400	800	1 200
Scrap/rework orders (Numbers of orders)	432 000	300	600	900
Product testing (Number of tests)	900 000	4 000	11 000	15 000
Machining (Number of machine hours)	2 500 000	20 000	30 000	50 000
Total Overhead Cost	\$4 000 000			

- Compute the activity application rate for EACH activity centre for 2009.
[4 marks]
- Compute the total overhead cost assigned to EACH product in 2009 using activity-based costing.
[4 marks]
- Compute the total cost to manufacture ONE unit of EACH product in 2009 using activity-based costing.
[2 marks]
- Based on your responses to a) through c) above, explain why the company's profitability has been declining since the introduction of the premium product.
[4 marks]

Total 35 marks

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3. (a) The president of Georgetown Company is evaluating three investment projects. The net cash flows for each project are estimated as follows:

Investment Project			
	A	B	C
Initial Cost	(\$212 508)	(\$227 448)	(\$217 180)
Year			
1	\$90 000	\$60 000	\$50 000
2	90 000	60 000	50 000
3	90 000	60 000	50 000
4		60 000	50 000
5		60 000	50 000
6			50 000
7			50 000
8			50 000

The firm's cost of capital is 12% and the following present value tables are available:

PVIF TABLE

Year	10%	11%	12%	13%	14%
0	1.0000	1.0000	1.0000	1.0000	1.0000
1	0.9091	0.9009	0.8929	0.8850	0.8772
2	0.8264	0.8116	0.7972	0.7831	0.7695
3	0.7513	0.7312	0.7118	0.6931	0.6759
4	0.6830	0.6587	0.6355	0.6133	0.5921
5	0.6209	0.5935	0.5674	0.5428	0.5194
6	0.5645	0.5346	0.5066	0.4803	0.4556
7	0.5132	0.4817	0.4523	0.4251	0.3996
8	0.4665	0.4339	0.4039	0.3762	0.3506
9	0.4241	0.3909	0.3606	0.3329	0.3075

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Compute EACH of the following:

- (i) The payback period for EACH investment project. [6 marks]
 - (ii) The net present value for EACH investment project. [15 marks]
 - (iii) The profitability index for EACH investment project. [3 marks]
- (b) Rank the three investment projects based on the three measures in (a) (i) – (iii). [3 marks]
- (c) Assume that investment project “B” involves the installation of equipment that will drastically reduce the emission of harmful gases from the company’s manufacturing plant. This plant is located in a pollution-ravaged city where the unemployment rate is higher than the national average.

Briefly describe FOUR non-financial benefits that may accrue to Georgetown Company if it decides to proceed with the investment in project “B”. [8 marks]

Total 35 marks

END OF TEST